



GE Medical Systems

Technical Publication

Direction 2337856-100

Revision 1

**LightSpeed Plus/Ultra V303
S/W Patch Load Instructions**

SERVICE DOCUMENTATION

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OMISSIONS & ERRORS

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IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- This Service Manual is available in English only.
- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this Warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

AVERTISSEMENT

- Ce Manuel de maintenance n'est disponible qu'en anglais.
- Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire.
- Ne pas tenter d'intervention sur les équipements tant que le manuel Service n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG

- Dieses Kundendienst-Handbuch existiert nur in englischer Sprache.
- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht, das Gerät zu reparieren, bevor dieses Kundendienst-Handbuch nicht zu Rate gezogen und verstanden wurde.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch elektrische Schläge, mechanische oder sonstige Gefahren kommen.

AVISO

- Este Manual de Servicio sólo existe en inglés.
- Si algún proveedor de servicios ajeno a GEMS solicita un idioma que no sea el inglés, es responsabilidad del cliente ofrecer un servicio de traducción.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

LANGUAGE

ATENÇÃO

- Este Manual de Assistência Técnica só se encontra disponível em Inglês.
- Se qualquer outro serviço de assistência técnica, que não a GEMS, solicitar estes manuais noutra língua, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este Manual de Assistência Técnica.
- O não cumprimento deste aviso pode por em perigo a segurança do técnico, operador ou paciente devido a choques elétricos, mecânicos ou outros.

AVVERTENZA

- Il presente manuale di manutenzione è disponibile soltanto in inglese.
- Se un addetto alla manutenzione esterno alla GEMS richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Non tenere conto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.

出口臺灣

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• 到随责良行自需与客
• 参本行抵数不，前之册于本编了全宗味数因股若未
• 通知数人录短员非数，员参本行会取事意主本参数
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DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write, "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage. Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins. 14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment. Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives. GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

REVISION HISTORY

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Revision	Date	Reason for change
0	3/29/02	Initial Release
1	07/02/02	CQA 1026195

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INSTALLATION

1.0 OVERVIEW

This software patch can be loaded via CD-ROM or via a software download from the Automated Support Center (ASC). The software includes fixes for LightSpeed Plus/Ultra software version 303 (H3_M4) filming issues and display slowdown.

These instructions include both the CD load instructions as well as the download instructions. The software download occurs in two separate steps. First, the software is downloaded to the scanner from the ASC. This does not install the patch; rather, it puts the software and installation script resident onto the scanner to be physically installed upon the next customer reboot. Once this download of the software package and install script has been successful, the customer will have to respond to a dialog box which will appear on their screen upon their next reboot, asking them if they wish to install the software at this time. If they answer YES, the software will install itself on the scanner and the scanner will automatically be rebooted to enable to changes. If the customer answers NO, the software will not be installed and the customer will not be prompted again to load the software. It is recommended that the customer be contacted and informed of the download prior to requesting it from the ASC so they are aware of the dialog box upon their next boot up and can respond appropriately.

This document also includes the instructions on how to remotely verify if the customer has rebooted the system and installed the software on the scanner (i.e. the customer answered YES to the dialog box asking them to load the software).

1.1 PREREQUISITES

LightSpeed Plus/Ultra Version 303 (H3_M4) software is required on this system prior to loading the V303 Patch software.

To verify that this software is loaded on the system, check *showprods*.

```
showprods | grep i Apps <ENTER>
```

```
<Apps ver 303.1_H3_M4
```

The above line shows the Product at Version 303 software. If the above line does not contain the string **Apps ver 303.1_H3_M4**, the scanner does not have the most recent H3_M4 software and FMI 25300 will have to be completed prior to implementing this software patch.

To verify that this patch is loaded on the scanner, please see section 3 of this document, *VERIFICATION of PATCH SOFTWARE LOAD* and section 6 of this document, *REMOTE VERIFICATION OF PATCH SOFTWARE LOAD*.

NOTICE: The above checks should be done if the patch software is being loaded on site. If the remote download is used, the download script performs the above checks automatically. If the incorrect version of software is detected, the download will not be completed and a message will be returned to the requestor.

2.0 INSTALLING VERSION 303 PATCH FROM CD

Shut down applications by:

Service -> Utilities -> Application Shutdown

Open a UNIX shell and become root.

type: **su -**

Insert Ver 303 patch CD in the CDROM Drive:

type: **mountCD /CDROM**

type: **/CDROM/install_Warp3_FMI**

You will see a message stating the following:

Preparing to install the patch...

Loading Warp FMI patch

When the installation is complete,

type: **cd**

type: **unmountCD**

Reboot the system

3.0 VERIFICATION OF SOFTWARE PATCH LOAD

To verify Patch is installed in OC:

type: **showprods | grep -i Warp3 _FMI**

The output will look similar to the following:

```
I Warp3_FMI          03/20/2002    Patch for Warp3 M4 FMI
I Warp3_FMI.patchimage_sw  03/20/2002    Patch for Warp3 M4 FMI
I Warp3_FMI.patchimage_sw.OCROOT  03/20/2002    Files in OCROOT
```

4.0 INSTALLING VER 303 PATCH VIA DOWNLOAD FROM ASC:

The installation of the software via download consists of two separate activities. The first activity is the download of the software file and installation scripts to the scanner. This does not automatically install the software, rather, downloads the patch software and installation script locally on the scanner for installation upon the next reboot. It is recommended that you contact your customer before initiating a download so they are aware of the software and installation dialog box.

4.1 INITIATING THE SOFTWARE AND INSTALL SCRIPT DOWNLOAD

To initiate the download of the software file and installation scripts to the scanner, send an e-mail as outlined below via Outlook or other mail tool. The message must be addressed properly and the subject line completed properly or your sweep will not be successful.

Message must have the following:

TO: **autosc (MED)** OR **autosc@olc.med.ge.com**
SUBJECT: **sweep**

Within the body of the email, the following must be typed exactly as it appears below:

```
type = warp3m4
system id = xxx
system id = yyy
.
.
system id = zzz
```

where XXX, YYY, ZZZ are the various system IDs that you wish to have loaded with the patch software.

4.2 CONFIRMING SUCCESSFUL DOWNLOAD

You will receive 2 messages from the ASC. The first will be a brief email confirming its receipt of your sweep (download) request and will look similar to the following:

Thank you for using the Automated Support Center!!

*Your request has been received by the Automated Support Center.
Data collected by the AutoSC will be returned to you in a separate mailing. Should we have trouble completing your request, you will*

instead receive a mailing listing the reasons your job could not be processed.

- - - - -
Note: Do you want more information about the status of your sweeps?
Check the AutoSC web page - <http://olcwebd.olc.med.ge.com/ss/asc/remotetouch>
We'd appreciate any comments - send email to autossc_help@med.ge.com
- - - - -

Request type: warp3m4
Request ID : 00095764
Requested by: Jane.Doe@med.ge.com
Respond to : Jane.Doe@med.ge.com
Num of System ID : 1

Please contact autossc_help@med.ge.com if you do not receive a response within one business day.

Questions and/or comments should be addressed to:
autossc_help@med.ge.com

GEMS Automated Support Center
*Dial Comm: 8*579-2800 Direct: (262) 896-2800*
Fax: (262) 896-2790 Email: autossc_help@med.ge.com
<http://3.7.192.186/ss/asc/Home.html>

The second will be a confirmation of the download of the software patch and installation script. The following is a typical example of an email confirmation to a successful Patch download.

PATCH was run. Please review for more information.

*Please do not reply directly to this message.
Questions and/or*

*comments should be addressed to:
asc_core@iscmed.med.ge.com*

*SYSTEM ID: CT_BAY6
USN: 0000CTBAY6
MLN: XXXX
Site Name: Ryerson Test System
Modality: CT
Product Line: HELIOS
Date: 2001:05:03:12:22:50*

*- - - - -
- - - - -*

*-----
Warp3 M4 FMI Patch was run on Thu Mar 28 11:40:45
CDT 2002*

*Hospital Name: G.E. Medical Systems
Service ID: CT_BAY6
Unique System ID: 0000CTBAY6*

patches:

*-----
x Attention.en_US, 396 bytes, 1 block
x Attention.es_MX, 466 bytes, 1 block
x Attention.fr, 446 bytes, 1 block
x Attention.it, 458 bytes, 1 block
x Attention.pt_BR, 459 bytes, 1 block
x PopupPatchInstall, 629 bytes, 2 blocks
x S99installWarp3, 48 bytes, 1 block
x StartPatchInstall, 1090 bytes, 3 blocks
x Warp3_FMI, 667 bytes, 2 blocks*

```
x Warp3_FMI.idb, 769 bytes, 2 blocks
x Warp3_FMI.patchimage_sw, 11703714 bytes, 22859 blocks
x disable, 2625 bytes, 6 blocks
x install_Warp3_FMI, 2039 bytes, 4 blocks
x uninstall_Warp3_FMI, 722 bytes, 2 blocks
```

Warp3 M4 FMI Patch successfully downloaded

Note:

It is possible that the first download attempt(s) by the ASC will not be successful. If this is the case, a failure message will be sent to you. The ASC will typically try three (3) times to download the software over a period of approximately 90 minutes although the actual response could be several hours, depending upon the ASC backlog, etc.

Note:

The downloaded script gets loaded to and runs from the /tmp directory. The support file, Warp3_FMI_patch.tar.gz by default, loads to the home dir of the account that the ASC logs in as. The ASC normally logs in as root, so Warp3_FMI_patch.tar.gz would be in /. If the ASC fails to log in as root, it will try as InSite, etc.. This means that Warp3_FMI_patch.tar.gz could end up in various locations, depending upon how the ASC successfully logs into the scanner. The file will be uncompressed and extracted under "/usr/g/dist" directory. One more script S99installWarp3 will be installed under /etc/rc2.d directory.

5.0 INSTALLING THE PATCH AFTER DOWNLOAD

At this point the download has been successful but the software has not been physically installed on the scanner. On the next reboot after the sweep has successfully downloaded the software to the scanner, the customer will be prompted with a question via a popup menu to accept the installation of the patch software (Y/N). If the customer selects YES, the installation script will be started automatically, the downloaded files will be installed on the scanner, and the system rebooted. It will take approximately 8 minutes for application software to be back up and ready for use.

If the customer selects NO, the patch will not be installed in the system, and another load (either download or load from CD) will be required. Again, it is recommended that the customer be made aware of the download prior to it happening so they can answer appropriately. One reason the customer may select NO is if the first reboot after the download occurs with a patient on the table and the customer cannot wait the additional 8 minutes required by the installation script. If they answer NO, they will not be prompted again and the software patch will have to be reloaded.

6.0 REMOTE VERIFICATION OF SOFTWARE PATCH LOAD

6.1 REQUESTING REMOTE CONFIRMATION OF CUSTOMER INSTALLATION

To remotely verify that the customer has installed the software on their next reboot, you will need to request another sweep as outlined below. This sweep should be run after you expect the customer to have installed the software, typically 24-48 hours after the successful download. The message must be addressed properly and the subject line completed properly or your sweep will not be successful.

Message must have the following:

TO: **autosc (MED)** OR **autosc@olc.med.ge.com**

SUBJECT: **sweep**

Within the body of the email, the following must be typed exactly as it appears below:

type = confwarp

system id = xxx

system id = yyy

.

.

system id = zzz

where XXX, YYY, ZZZ are the various system IDs that you wish to verify have installed the patch on their next reboot.

6.2 CONFIRMATION OF CUSTOMER INSTALLATION (REBOOT)

You will receive 2 messages from the ASC. The first will be a brief email confirming its receipt of your sweep (confirmation) request and will look similar to the following:

Thank you for using the Automated Support Center!!

*Your request has been received by the Automated Support Center.
Data collected by the AutoSC will be returned to you in a separate mailing. Should we have trouble completing your request, you will instead receive a mailing listing the reasons your job could not be processed.*

- - - - -
*Note: Do you want more information about the status of your sweeps?
Check the AutoSC web page - <http://olcwebd.olc.med.ge.com/ss/asc/remotetouch>
We'd appreciate any comments - send email to autossc_help@med.ge.com*
- - - - -

*Request type: confwarp
Request ID : 00100230
Requested by: Jane.Doe@med.ge.com
Respond to : Jane.Doe@med.ge.com
Num of System ID : 1*

*Please contact autossc_help@med.ge.com if you do not receive a
response within one business day.*

*Questions and/or comments should be addressed to:
autossc_help@med.ge.com*

*GEMS Automated Support Center
Dial Comm: 8*579-2800 Direct: (262) 896-2800
Fax: (262) 896-2790 Email: autossc_help@med.ge.com
<http://3.7.192.186/ss/asc/Home.html>*

The second will be a confirmation of the installation of the software patch indicating the customer rebooted the system and answered YES to the pop-up window. The following is a typical example of an email confirmation to a successful Patch installed on the scanner:

CONFIRMWARP3PATCH was run. Please review for more information.

*Please do not reply directly to this message. Questions and/or
comments should be addressed to: asc_core@iscmed.med.ge.com*

SYSTEM ID: 0000CTBAY6
USN: 0000359045
MLN: 9998
Site Name: ST Anywhere HOSPITAL
Modality: CT
Product Line: HELIOS
Date: 2001:06:12:13:02:05

- - - - -

Hospital Name: ST Anywhere HOSPITAL, AZ 12

Service ID: 12345CTQ

Unique System ID: 0000CTBAY6

03/29/2002 Patch for Warp3 M4 FMI

patches:

I Warp3_FMI.patchimage_sw 03/29/2002 Patch for Warp3 M4 FMI

I Warp3_FMI.patchimage_sw.OCR00T 03/29/2002 Files in OCR00T

Warp3 M4 FMI Patch exist on the system & was installed on 03/29/2002

Exiting...

- - - - -

7.0 UNINSTALLATION OF VERSION 303 PATCH

=====

Shutdown Applications by:

Service -> Utilities -> Application Shutdown

Become root by typing: **su -**

type: **/usr/g/scripts/uninstall_Warp3_FMI**

Restart applications by typing **St**



GE Medical Systems

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